



Basic Input and Output in Java

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Lecture #	5
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Week #	2

Interacting with a Java program

- We have seen how to print data
 - `System.out.println("Hello, world");`
- But how do we read data?

Reading input

- Simplest to use is the `Console` class
 - Functionality quite similar to Python `input()`
- Defined within `System`
 - Two methods, `readLine` and `readPassword`
 - `readPassword` does not echo characters on the screen
 - `readLine` returns a string (like Python `input()`)
 - `readPassword` returns an array of `char`

```
Console cons = System.console();
String username = cons.readLine("Username: ");
char[] password = cons.readPassword("Password: ");
```

- A more general `Scanner` class
 - Allows more granular reading of the input
 - Read a full line, or read an integer ...

```
Scanner in = new Scanner(System.in);
String name = in.nextLine();
int age = in.nextInt();
```

Generating Output

- `System.out.println(arg)` prints `arg` and goes to a new line
 - Implicitly converts argument to a string
- `System.out.print(arg)` is similar, but does not advance to a new line
- `System.out.printf(arg)` generates formatted output
 - Same conventions as `printf` in C language